CDAC ADMISSION CASE STUDY

Create tables and insert in tables for : ccat_result , centers, courses , center_courses , preferences.

Command Prompt - mysql -u KD3_Siddharth_96182 -p

```
KD3_Siddharth_96182>CREATE DATABASE cdac_admission_system;
Query OK, 1 row affected (0.01 sec)

KD3_Siddharth_96182>USE cdac_admission_system;
Database changed
KD3_Siddharth_96182>CREATE TABLE ccat_results (formno INT PRIMARY KEY,name VARCHAR(50),c_rank INT);
Query OK, 0 rows affected (0.03 sec)

KD3_Siddharth_96182>INSERT INTO ccat_results VALUES
    -> (1001,'Amit',120),
    -> (1002,'Rahul',90),
    -> (1003,'Sneha',150),
    -> (1004,'Pooja',110);
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0

KD3_Siddharth_96182>CREATE TABLE centers (
    ->     center_id INT PRIMARY KEY,
    ->     center_name VARCHAR(50),
    ->     city VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.03 sec)

KD3_Siddharth_96182>INSERT INTO centers VALUES
    -> (1,'Sunbeam','Pune'),
    -> (2,'ACTS','Pune'),
    -> (3,'KnowIT','Pune'),
    -> (4,'Kharghar','Mumbai');
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0

KD3_Siddharth_96182>
```

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Records: 4  Duplicates: 0  Warnings: 0

KD3_Siddharth_96182>CREATE TABLE courses (
    ->     course_id INT PRIMARY KEY,
    ->     course_name VARCHAR(50),
    ->     duration VARCHAR(20)
    -> );
Query OK, 0 rows affected (0.04 sec)

KD3_Siddharth_96182>INSERT INTO courses VALUES
    -> (1,'PG-DAC','6 Months'),
    -> (2,'PG-DBDA','6 Months'),
    -> (3,'PG-DESD','6 Months');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

KD3_Siddharth_96182>CREATE TABLE center_courses (
    ->     center_course_id INT AUTO_INCREMENT PRIMARY KEY,
    ->     center_id INT,
    ->     course_id INT,
    ->     seats INT,
    ->     FOREIGN KEY (center_id) REFERENCES centers(center_id),
    ->     FOREIGN KEY (course_id) REFERENCES courses(course_id)
    -> );
Query OK, 0 rows affected (0.54 sec)

KD3_Siddharth_96182>INSERT INTO center_courses (center_id,course_id,seats) VALUES
    -> (1,1,2),
    -> (1,2,1),
    -> (2,1,2),
    -> (3,2,2),
    -> (4,3,2);
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

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```
KD3_Siddharth_96182>CREATE TABLE preferences (
    ->     preference_id INT AUTO_INCREMENT PRIMARY KEY,
    ->     formno INT,
    ->     center_course_id INT,
    ->     preference_order INT,
    ->     FOREIGN KEY (formno) REFERENCES ccat_results(formno),
    ->     FOREIGN KEY (center_course_id) REFERENCES center_courses(center_course_id)
    -> );
Query OK, 0 rows affected (0.07 sec)

KD3_Siddharth_96182>CREATE TABLE allocations (
    ->     allocation_id INT AUTO_INCREMENT PRIMARY KEY,
    ->     formno INT,
    ->     center_course_id INT,
    ->     round_no INT,
    ->     FOREIGN KEY (formno) REFERENCES ccat_results(formno),
    ->     FOREIGN KEY (center_course_id) REFERENCES center_courses(center_course_id)
    -> );
Query OK, 0 rows affected (0.14 sec)

KD3_Siddharth_96182>DELIMITER $$
KD3_Siddharth_96182>
KD3_Siddharth_96182>CREATE PROCEDURE allocate_round(IN round_number INT)
    -> BEGIN
    ->
    -> DECLARE done INT DEFAULT 0;
    -> DECLARE f_no INT;
    -> DECLARE cc_id INT;
    -> DECLARE pref INT;
    -> DECLARE seat_count INT;
    ->
    -> DECLARE student_cursor CURSOR FOR
    -> SELECT formno
    -> FROM ccat_results
```

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```
KD3_Siddharth_96182>DELIMITER $$
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KD3_Siddharth_96182>CREATE PROCEDURE allocate_round(IN round_number INT)
    -> BEGIN
    ->
    -> DECLARE done INT DEFAULT 0;
    -> DECLARE f_no INT;
    -> DECLARE cc_id INT;
    -> DECLARE pref INT;
    -> DECLARE seat_count INT;
    ->
    -> DECLARE student_cursor CURSOR FOR
    -> SELECT formno
    -> FROM ccat_results
    -> ORDER BY c_rank ASC;
    ->
    -> DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = 1;
    ->
    -> OPEN student_cursor;
    ->
    -> student_loop: LOOP
    ->
    -> FETCH student_cursor INTO f_no;
    ->
    -> IF done = 1 THEN
    ->     LEAVE student_loop;
    -> END IF;
    ->
    -> SET pref = 1;
    ->
    -> preference_loop: LOOP
    ->
    -> SELECT center_course_id INTO cc_id
    -> FROM preferences
```

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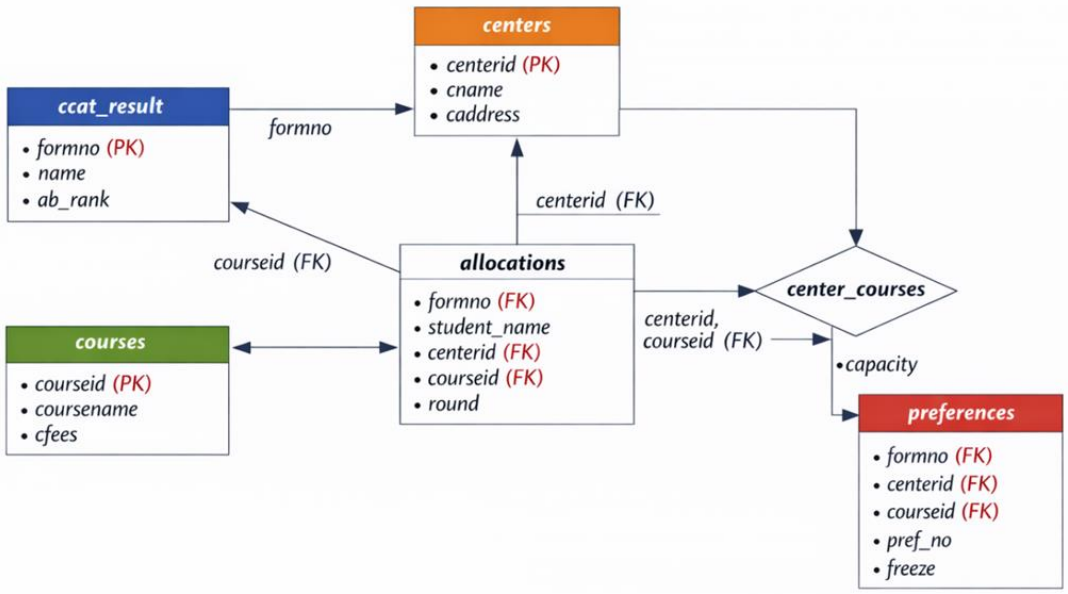
```
    -> WHERE formno = f_no AND preference_order = pref;
    ->
    -> IF cc_id IS NULL THEN
    ->     LEAVE preference_loop;
    -> END IF;
    ->
    -> SELECT seats INTO seat_count
    -> FROM center_courses
    -> WHERE center_course_id = cc_id;
    ->
    -> IF seat_count > 0 THEN
    ->
    ->     INSERT INTO allocations(formno, center_course_id, round_no)
    ->     VALUES (f_no, cc_id, round_number);
    ->
    ->     UPDATE center_courses
    ->     SET seats = seats - 1
    ->     WHERE center_course_id = cc_id;
    ->
    ->     LEAVE preference_loop;
    ->
    -> END IF;
    ->
    -> SET pref = pref + 1;
    ->
    -> END LOOP;
    ->
    -> END LOOP;
    ->
    -> CLOSE student_cursor;
    ->
    -> END$$
Query OK, 0 rows affected (0.02 sec)

KD3_Siddharth_96182>
```

```
KD3_Siddharth_96182>CALL allocate_round(1);
Query OK, 0 rows affected (0.03 sec)

KD3_Siddharth_96182>CALL allocate_round(2);
Query OK, 0 rows affected (0.01 sec)

KD3_Siddharth_96182>SELECT
    -> c.formno,
    -> c.name,
    -> c.c_rank,
    -> ce.center_name,
    -> co.course_name,
    -> a.round_no
    -> FROM allocations a
    -> JOIN ccat_results c ON a.formno = c.formno
    -> JOIN center_courses cc ON a.center_course_id = cc.center_course_id
    -> JOIN centers ce ON cc.center_id = ce.center_id
    -> JOIN courses co ON cc.course_id = co.course_id
    -> ORDER BY c.c_rank;
Empty set (0.01 sec)

KD3_Siddharth_96182>
```

CDAC Admission System – ER Diagram